NVIDIA VSP integration — issues and fixes

Date: 2026-10-25. Work is uncommitted on branch `LifecycleManager`. Nothing below was committed in this stretch.

Summary

Issue	Result
Daemon and VSP did not share a socket	Fixed in manifests. Live handshake proved on dh2
SoC management `0xc2d3` could look like a second DPU	Fixed. Function 0 BlueField products only
Missing PCI or serial could still stamp a DPU	Fixed. `ValidateIdentity` blocks the stamp
Stale annotations survived reboot, removal, and a second card	Fixed. Inventory refresh clears or rebinds from a live scan
CreateNetworkFunction succeeded before DPF accepted anything	Fixed. Success requires a live chart-bearing DPUService
Kind `make test-e2e` never ran the new image	Open. Docker socket down and disk was full
dh2 has no `go` binary	Left as-is. Policy forbids installing a toolchain
`DPU/dh2-bluefield-2` stuck in Rebooting	Open. Flavor cannot be recreated while the DPU refers to it
Override annotations made the manager error-loop	Fixed on the cluster by removing the seven keys. Flavor apply succeeded again

1. Socket was not the daemon socket

Kind used an emptyDir the official daemon cannot see. Hardware needs the host path the dpu-operator already mounts.

Fix: `config/vsp/daemonset.yaml` shares an emptyDir with an `opi-daemon` sidecar. Overlay `config/overlays/hardware-vsp` uses hostPath `/var/run/dpu-daemon` (Bidirectional) and drops the sidecar. `cmd/opi-daemon` is the GrpcPlugin Init/GetDevices client. Directory mode is `0700`. Default socket is `/var/run/dpu-daemon/vendor-plugin/vendor-plugin.sock`.

Live proof: VSP pod `opi-nvidia-dpf-adapter-nvidia-vsp-bfhws` Running. `opi-daemon --once` logged one Healthy device.

2. PCI identity was too loose

An NVIDIA fallback treated the SoC management function `0xc2d3` as a DPU. On dh2 that is `0000:0f:00.2`, next to the real Ethernet functions `.0` and `.1` (`0x15b3:0xa2d6`, MT42822). sysfs `serial` is missing. VPD and config space are readable as root.

Fix in `pkg/discovery/pci.go`: products are BlueField-2 `0xa2d2` / `0xa2d6` and BlueField-3 `0xa2dc` / `0xa2dd`. No fallback. `uniqueBySerial` keeps one function 0 device per serial. Serial read order is sysfs `serial`, then VPD `SN`, then the PCIe DSN. Permission denied is a hard error.

Live identity: PCI `0000:0f:00.0`, DPF form `0000-0f-00`, serial `MT2321XZ0KVX`. `.1` is the same serial and is dropped. `.2` is skipped.

3. Missing identity still looked provisionable

A DPU with an empty PCI address or serial could pass the mapping.

Fix: `ValidateIdentity` requires both. Mapping fields `spec.serialNumber` and `spec.pciAddress` are `required: true`. The annotator does not stamp an invalid device. `TestApplyBlocksDPUWithoutPCIAddress` covers the mapping.

4. Reboot, disappearance, and a second card

Annotations were treated as proof of hardware. After a rescan the PCI address can move. A stale serial could steal another DPU's card. An enumerate error used to look like an empty bus.

Fix: `pkg/discovery/inventory.go`. Refresh every 15s. `LiveDevices` drops invalid identities and returns the enumerate error instead of clearing. `AssignDevices` matches serial, then PCI, then unused cards onto unstamped DPUs (name order against lowest PCI). A stale serial is cleared and cannot steal. Two stale DPUs plus one leftover card are not guessed. One leftover card may replace one cleared DPU. `StripIdentity` removes serial and PCI only. Firmware URL and flavor stay.

Tests: idempotent bind, clear when gone, PCI move, PCI swap, multi-device pair, two-stale-one-card, enumerate failure keeps stamps, annotator follows a hardware change and ignores the other DPU.

5. Network-function success was a no-op

Create and Delete returned an empty success before any DPUService existed. That reported a user-facing success with no DPF intent.

Fix: `pkg/vsp/handoff.go`. Selected contract is delegate-to-translator/DPF (DPF 26.4 DPUService guide). The VSP does not Helm-install, Helm-uninstall, or mutate DPF. Create succeeds only when a DPUService exists, is not deleting, and `spec.helmChart.source.chart` is set. Delete is CNI detach: it succeeds when DPF is observable, and a live DPUService is not a delete failure. Nil handoff is `RejectHandoff`. In-cluster VSP sets `NewClusterHandoff`. RBAC on `svc.dpu.nvidia.com/dpuservices` is get, list, and watch only.

Tests: refuse before intent, static handoff, fake client empty / accepted / empty chart / deleting, nil client, RBAC file has no create/update/patch/delete, doc string contains `do not Helm-install` on one line. Live dh2: 11 accepted DPUServices, create error nil.

6. Kind end-to-end image

`make test-e2e` must use an image that contains `/opi-daemon`. The run failed before the cluster existed:

```
failed to connect to the docker API at
unix:///Users/kartikey0104/.docker/run/docker.sock
make: *** [setup-test-e2e] Error 1
```

The Mac data volume had also been full, so the image was not rebuilt. This is still open. Unit and lab hardware tests do not depend on Kind.

7. Lab test packaging

Mapping tests failed on `._amd-dsc200.yaml` (AppleDouble) and on a tree that was missing `config/condition-writers.yaml`, CRD fixtures, and sibling Go sources. `cmd/opi-daemon` and `cmd/vsp` tests call `go build` and fail on dh2 with `exec: "go": executable file not found`. Conformance BeforeSuite skips without envtest binaries. The permission-denied PCI test skips when euid is 0.

Fix for the YAML failures: `COPYFILE_DISABLE=1`, delete `._*`, and upload a repo-shaped tree under `/tmp/lab-job-<name>/`. Go was not installed on dh2.

8. Reboot is stuck after a successful install

`DPU/dh2-bluefield-2` phase is `Rebooting`, previous phase `DPU Config`. OSInstalled, BFBReady, and FWConfigured are True. OperationalReady is Unknown because the node has not joined. Host agent at 2026-10-25T05:38:00Z: `NVConfigApplied=True`, `RebootMethodDiscovery=True/SystemLevelReset`, `reset_needed=true`, description `PCI rescan is required`, command `Reboot external host is required`. Running firmware `24.49.1014`. Scale functions already match (`PF_TOTAL_SF=32`, `PF_SF_BAR_SIZE=8`). Still pending next boot:

Parameter	Current	Next boot
LAG_RESOURCE_ALLOCATION	0	1

Parameter	Current	Next boot
NUM_OF_VFS	8	46
NUM_PF_MSIX_VALID	1	0
PER_PF_NUM_SF	0	1
PF_BAR2_ENABLE	1	0
PF_NUM_PF_MSIX	63	228
PF_NUM_PF_MSIX_VALID	0	1

Code fix for a future flavor: `config/mappings/dataprocessingunit.yaml` can override those parameters from annotations. `0` on the two scale-function keys still falls back to official 20 and 10. Other parameters keep an explicit 0. `TestApplyFlavorNvconfigMatchesRunningFirmware` passed locally with the dh2 current values.

That fix cannot land on the live flavor. DPF `spec` is immutable. Deleting `dpf-default-flavor` was denied by `vdpuflavor.kb.io`: the flavor is referred to by `DPU/dh2-bluefield-2`, and the DPU must be deleted first. The DPU finalizer `provisioning.dpu.nvidia.com/dpu-protection` deprovisions the card, so the DPU was not deleted. `mlxfwreset` was not run. The host was not rebooted.

Setting the seven current-firmware annotations made the controller report `DPUFlavor spec is immutable` on every reconcile. Those seven annotations were removed. `dpu.nvidia.com/pf-total-sf=32` and `pf-sf-bar-size=8` stayed, because they already match the flavor. The next reconciles logged `applied translated object` for `DPUFlavor/dpf-default-flavor`. Phase is still Rebooting.

9. Prerequisites that were checked and left alone

Present and not reinstalled: cert-manager controller (newer than the v26.4.0 doc pin v1.19.3), Argo CD (newer than chart 9.4.1), node-feature-discovery, maintenance-operator `v0.3.0`, DPF system `v26.4.0`. Kamaji is absent. `DPUCluster/dpu-cluster-1` is static and Ready, so Kamaji is not required. Unhealthy flannel, cni-installer, sfc, and sriov pods follow from the DPU node not joining. They were not treated as a separate install.